

OSSS 66

Question:

To Create Process Escape Sequences, Handle Escaped Spaces, Escape Special Symbols, Preserve Characters, Validate Parser Output, Test Complex Inputs.

To Create Child Processes, Execute Programs, Handle Execution Errors, Pass Arguments, Manage Parent Process, Test Command Launching.

Solution:

```
ubuntu@ubuntu:~$ nano escape_parser.c
ubuntu@ubuntu:~$ gcc escape_parser.c -o escape_parser
ubuntu@ubuntu:~$ ./escape_parser
Enter Command: Hello\ World

Parsed Output:
Hello World
```

```
GNU nano 8.7.1 escape_parser.c
#include <stdio.h>
#include <string.h>

int main()
{
    char input[200];
    printf("Enter Command: ");
    fgets(input, sizeof(input), stdin);

    printf("\nParsed Output:\n");

    for(int i = 0; input[i] != '\0'; i++)
    {
        if(input[i] == '\\')
        {
            i++;
            if(input[i] != '\0')
                printf("%c", input[i]);
        }
        else
        {
            printf("%c", input[i]);
        }
    }

    return 0;
}
```

```
ubuntu@ubuntu:~$ nano process_launcher.c
ubuntu@ubuntu:~$ gcc process_launcher.c -o process_launcher
ubuntu@ubuntu:~$ ./process_launcher
Child Process PID = 6296
total 40
drwxr-xr-x 2 ubuntu ubuntu 60 Sep 30 02:50 Desktop
drwxr-xr-x 2 ubuntu ubuntu 40 Sep 30 02:51 Documents
drwxr-xr-x 2 ubuntu ubuntu 40 Sep 30 02:51 Downloads
drwxr-xr-x 2 ubuntu ubuntu 40 Sep 30 02:51 Music
drwxr-xr-x 2 ubuntu ubuntu 40 Sep 30 02:51 Pictures
drwxr-xr-x 2 ubuntu ubuntu 40 Sep 30 02:51 Public
drwxr-xr-x 2 ubuntu ubuntu 40 Sep 30 02:51 Templates
drwxr-xr-x 2 ubuntu ubuntu 40 Sep 30 02:51 Videos
-rwxrwxr-x 1 ubuntu ubuntu 70728 Sep 30 02:56 escape_parser
-rw-rw-r-- 1 ubuntu ubuntu 462 Sep 30 02:55 escape_parser.c
-rwxrwxr-x 1 ubuntu ubuntu 70752 Sep 30 02:58 process_launcher
-rw-rw-r-- 1 ubuntu ubuntu 553 Sep 30 02:58 process_launcher.c
drwx----- 5 ubuntu ubuntu 100 Sep 30 02:51 snap

Parent Process PID = 6295
Child Completed Successfully
```

```

#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
#include <stdlib.h>

int main()
{
    pid_t pid;

    pid = fork();

    if(pid < 0)
    {
        perror("Fork Failed");
        return 1;
    }

    if(pid == 0)
    {
        printf("Child Process PID = %d\n", getpid());

        execlp("ls", "ls", "-l", NULL);

        perror("Execution Failed");
        exit(1);
    }
    else
    {
        wait(NULL);

        printf("\nParent Process PID = %d\n", getpid());

        printf("Child Completed Successfully\n");
    }
}

```

```

ubuntu@ubuntu:~$ nano process_launcher.c
ubuntu@ubuntu:~$ nano exec_args.c
ubuntu@ubuntu:~$ gcc exec_args.c -o exec_args
ubuntu@ubuntu:~$ ./exec_args
Launching grep...
int main()
    "main",

```

```

#include <stdio.h>
#include <unistd.h>

int main()
{
    printf("Launching grep...\n");

    execlp("grep",
           "grep",
           "main",
           "exec_args.c",
           NULL);

    return 0;
}

```